Eduqas GCSE Geography (C/4) — Component 1

1.1 Coastal Landscapes

1. The Coastal System

Coastlines are **open systems** — they receive inputs of energy and material, process them through various flows, and produce outputs. Understanding the coast as a system helps explain why coastlines constantly change.

Inputs

- **Wave energy** — the primary driver of coastal processes; generated by wind acting on the sea surface
- **Sediment** — supplied by rivers, cliff erosion, offshore sources, and wind (aeolian)
- **Solar energy** — drives evaporation, wind generation, and biological processes
- **Tidal energy** — tidal range determines how much of the beach/cliff is exposed to wave attack

Throughputs (flows and stores)

- **Erosion** — rock and sediment removed from cliffs and the sea bed
- **Transportation** — movement of sediment along and across the shore
- **Deposition** — sediment dropped when energy decreases
- **Stores** — beaches, dunes, and offshore bars act as temporary sediment stores

Outputs

- Sediment transported offshore or out of the system
- Heat energy dissipated by wave breaking
- Landforms (cliffs, beaches, spits) as long-term stores

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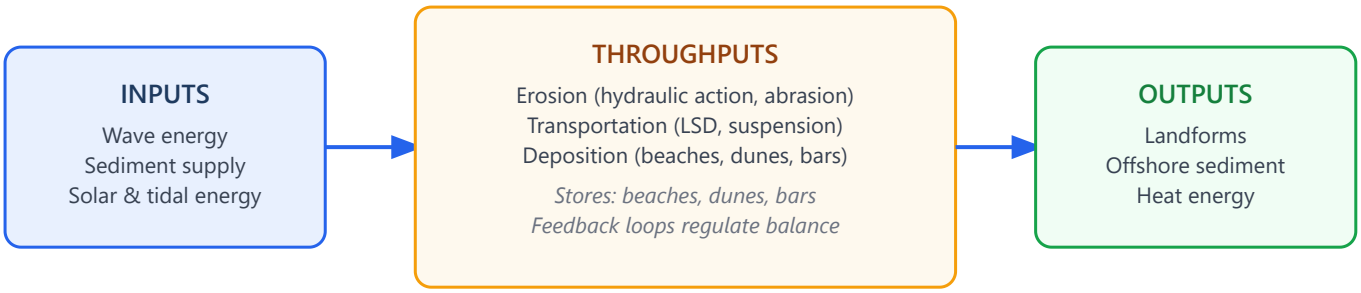

Fig 1.1 — The coastal system: inputs, throughputs, and outputs

Dynamic Equilibrium and Feedback

A coast in **dynamic equilibrium** maintains a balance between erosion and deposition over time. When this balance is disturbed (e.g., by a storm or sea-level rise), feedback mechanisms operate:

- **Negative feedback** — restores equilibrium. Example: cliff erosion provides more sediment, which builds a beach, which absorbs wave energy, reducing further erosion.
- **Positive feedback** — amplifies change. Example: rising sea levels increase wave energy at cliffs, increasing erosion, releasing more sediment, which is washed offshore, leaving cliffs more exposed — further accelerating retreat.

Exam Tip: When asked about the coastal system, use the words "inputs", "throughputs", "outputs", "stores", and "feedback". Examiners award marks for correct use of systems terminology.

2. Waves

Wave Formation

Waves are generated by wind transferring energy to the water surface. Three factors determine wave energy:

- **Wind speed** — faster winds generate larger waves
- **Wind duration** — longer sustained winds build bigger waves
- **Fetch** — the distance of open water over which wind blows. A long fetch (e.g., Atlantic waves reaching Cornwall) produces high-energy waves

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Water particles move in circular **orbital motion** — they return to roughly their starting position as each wave passes. It is energy, not water, that moves forward. When a wave reaches shallow water, friction slows the base of the wave, the top topples forward and the wave **breaks**.

Constructive vs Destructive Waves

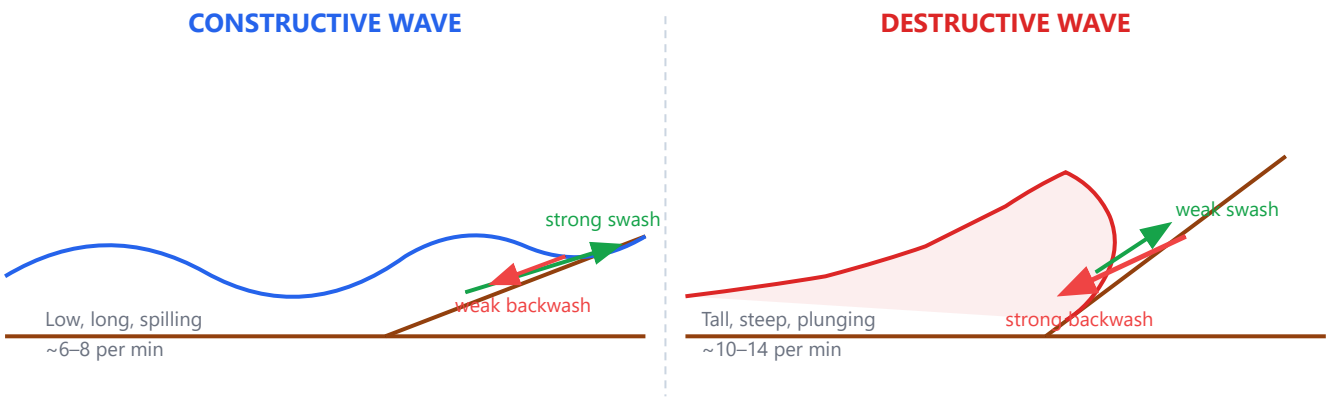

Fig 1.2 — Constructive (left) vs Destructive (right) waves

Feature	Constructive Wave	Destructive Wave
Height	Low (<1 m)	High (>1 m)
Frequency	6–8 per minute	10–14 per minute
Fetch	Short to moderate	Long (e.g., Atlantic)
Swash	Strong — pushes sediment up beach	Weak
Backwash	Weak — little material dragged back	Strong — pulls sediment offshore
Net effect	Beach building (deposition)	Beach erosion
Wave type	Spilling (breaks gently)	Plunging (curls and crashes)

Wave Refraction

When waves approach a coastline at an angle, the part of the wave in shallower water slows down first while the offshore part continues at speed. This bends (**refracts**) the wave so it becomes more parallel to

the shore. At headlands, waves wrap around and concentrate energy; at bays, energy is dispersed. This explains why headlands erode faster and bays accumulate sediment.

Exam Tip: For a 4–6 mark question on waves, always contrast constructive and destructive waves using specific data (frequency, height) and explain the net effect on the beach.

3. Coastal Erosion Processes

Hydraulic Action: The sheer force of waves crashing against a cliff face compresses air trapped in cracks and joints. When the wave retreats, pressure is suddenly released, causing an explosive effect that widens and extends cracks. Over time this shatters rock.

Abrasion (Corrasion): Waves pick up pebbles, sand, and boulders and hurl them at cliff faces, acting like sandpaper. This is the most effective erosion process and is responsible for the smooth wave-cut notch at the base of cliffs.

Attrition: Boulders and pebbles carried by waves collide with each other, gradually becoming smaller, rounder, and smoother. Attrition does not erode the cliff but reduces the size of beach sediment over time.

Corrosion (Solution): Weak carbonic acid in seawater slowly dissolves calcium carbonate in rocks such as limestone and chalk. It is a chemical process and operates slowly, but is important on carbonate coastlines.

Exam Tip: Do not confuse "abrasion" (waves using load to scrape cliffs) with "attrition" (load particles wearing each other down). Also note "corrosion" in geography means chemical dissolution — different from the everyday use.

4. Coastal Erosion Landforms

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Cliffs and Wave-Cut Platforms

Cliff formation occurs where hard rock meets the sea. Waves attack the base of the cliff through abrasion and hydraulic action, creating a **wave-cut notch**. As the notch deepens, the cliff above becomes unsupported and collapses. The cliff retreats landward, leaving behind a gently sloping **wave-cut platform** (also called an abrasion platform) exposed at low tide.

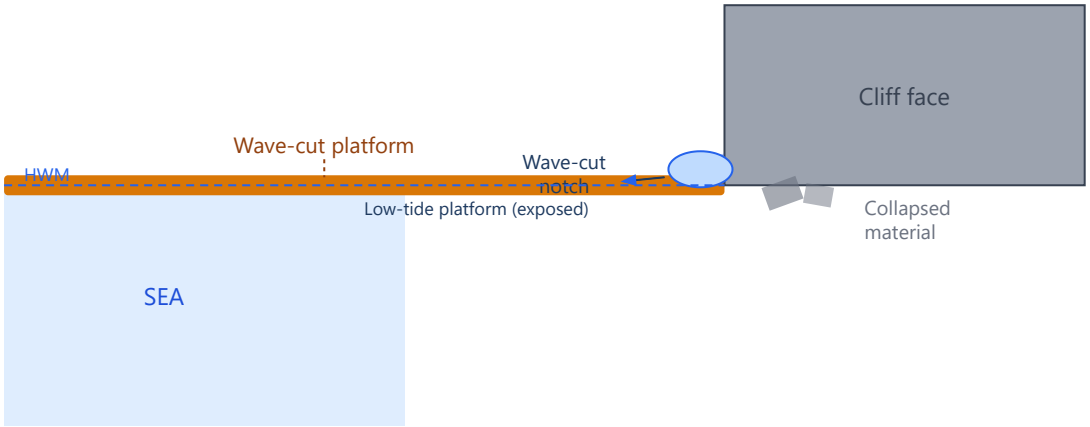

Fig 1.3 — Cross-section of a wave-cut notch and wave-cut platform

Headlands and Bays

Where a coastline has alternating bands of **hard and soft rock** running perpendicular to the sea (discordant geology), the soft rock erodes faster to form **bays** while the resistant rock protrudes as **headlands**. Wave refraction concentrates energy on headlands, accelerating their erosion. Bays receive lower-energy waves and accumulate beach sediment.

Example: Swanage Bay, Dorset — soft Wealden clays form the bay; hard Purbeck limestone and chalk form the headlands (Old Harry Rocks to the east, Ballard Point).

Caves, Arches, Stacks, and Stumps

These landforms represent a **sequence of erosion** at headlands:

1. **Cave** — hydraulic action and abrasion exploit a joint or weakness in a headland cliff face, creating a cave
2. **Arch** — if caves develop on opposite sides of a headland, they may join to form an arch through the headland
3. **Stack** — the roof of the arch collapses (gravity, weathering, wave attack), leaving an isolated pillar of rock called a stack

4. **Stump** — further erosion at the base of the stack causes it to collapse to a low, wave-cut stump, visible only at low tide

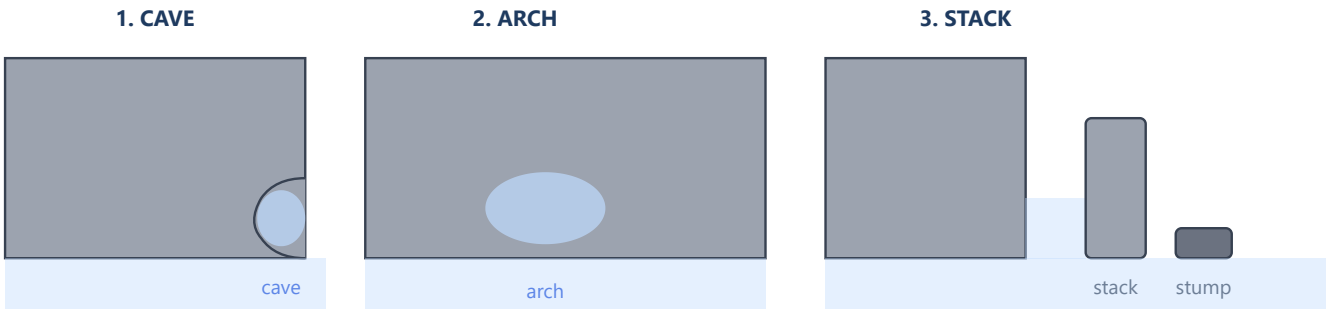

Fig 1.4 — Sequence of cave, arch, stack, and stump formation (e.g., Old Harry Rocks, Dorset)

Example: Old Harry Rocks, Dorset — chalk headland with a surviving stack (Old Harry) and a nearby stump (Old Harry's Wife). The Needles, Isle of Wight are chalk stacks.

5. Coastal Transportation

Sediment is moved along the coast in four main ways:

Traction: Large boulders and cobbles rolled along the sea bed by wave energy.

Saltation: Sand and small pebbles bounced along the sea bed in a series of leaps.

Suspension: Fine silt and clay particles carried within the water column — gives seawater its brown/murky colour near estuaries.

Solution: Dissolved minerals (e.g., calcium carbonate from limestone) transported invisibly in the water.

Longshore Drift (LSD)

This is the dominant mechanism for moving sediment along the coast. Waves rarely approach the shore perfectly parallel — they usually arrive at a slight angle. This causes sediment to move in a **zig-zag** pattern along the beach:

1. **Swash** pushes sediment up the beach at the same angle as the approaching wave
2. **Backwash** pulls sediment straight back down the beach under gravity (perpendicular to shore)
3. Net movement is in the direction of the prevailing wind/wave approach

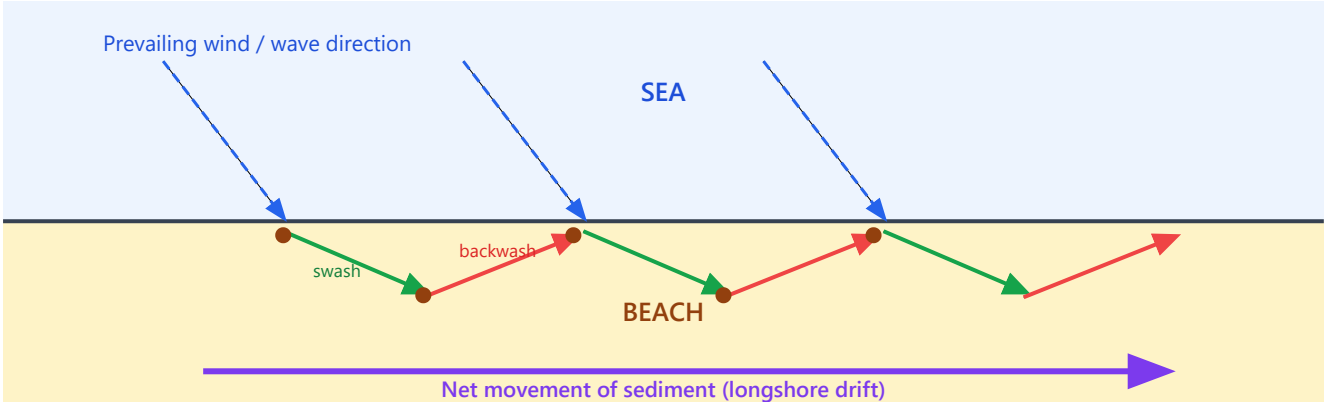

Fig 1.5 — Longshore drift mechanism showing swash (green), backwash (red), and net sediment movement (purple)

Exam Tip: Many coastal management features (groynes, breakwaters) exist to disrupt longshore drift. Always link back to LSD when explaining why these structures are built.

6. Coastal Deposition Landforms

Deposition occurs when wave energy decreases — for example, in sheltered bays, in the lee of a headland, or where a river estuary slows water movement.

Beaches

Beaches are accumulations of sediment (sand, shingle, pebbles) between the low and high water marks. Two types based on how they form:

- **Swash-aligned beaches** — waves arrive parallel to shore; no net LSD; sediment builds up in place (e.g., at the head of a bay)
- **Drift-aligned beaches** — waves arrive at an angle; LSD transports sediment along the shore; beach has a net movement of material

Beach profiles vary: **shingle beaches** are steep (steep swash, sediment doesn't travel far); **sand beaches** are gentle (water percolates through, reducing backwash).

Spits

A spit is a long, narrow ridge of sand or shingle extending from the land into the sea. It forms where the coastline suddenly changes direction (e.g., at a river mouth or a bay), causing LSD to deposit sediment in open water. The end of the spit may curve inland as secondary waves arriving from a different direction refract around the tip, forming a **recurved end**. Behind the spit, a sheltered lagoon or salt marsh develops.

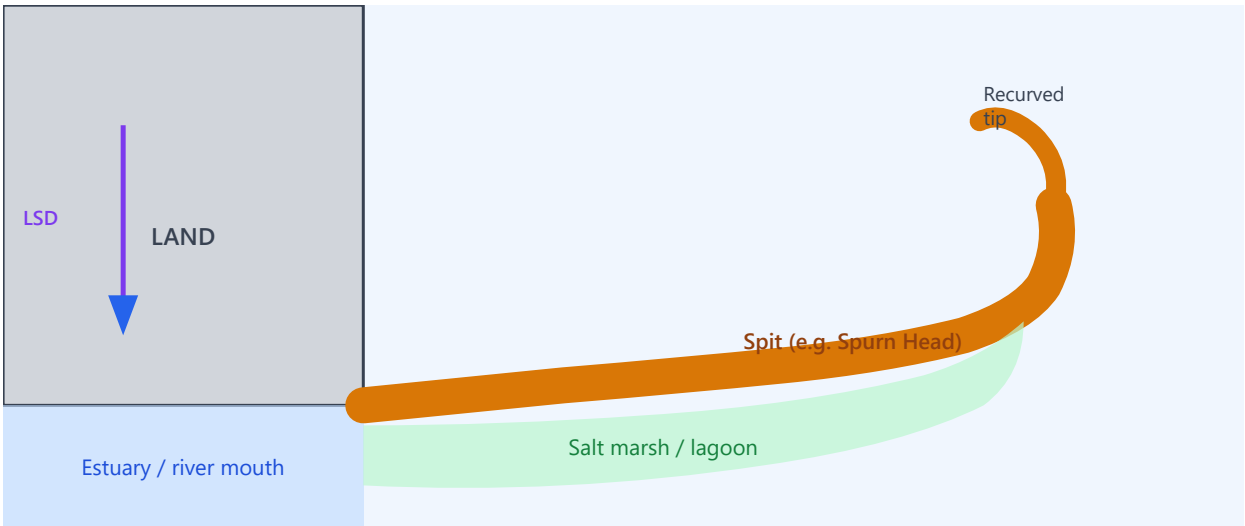

Fig 1.6 — Spit formation at a river mouth (e.g., Spurn Head, Humber Estuary, East Yorkshire)

Spurn Head, East Yorkshire — a 5.5 km curved spit extending into the Humber Estuary. Formed by dominant northerly LSD carrying sand and shingle south. The recurved tip is shaped by wave refraction from the south-east. It is 50–100 m wide in places and at severe risk from erosion.

Bars and Tombolos

- **Bar** — if a spit grows across a bay and joins the headlands on both sides, it creates a bay bar, enclosing a lagoon. Example: Slapton Sands, Devon.
- **Tombolo** — a bar that connects the mainland to an offshore island. Example: Chesil Beach connects mainland Dorset to the Isle of Portland.

Sand Dunes — Succession

Sand dunes form where sand is blown onshore and trapped by obstacles. Vegetation colonises dunes in a **succession**:

Stage	Dune Type	Characteristics	Key Plant

1	Embryo dunes	Tiny, unstable; form around strand-line debris; highly mobile; pH ~8 (alkaline)	Sea couch grass, sea rocket
2	Yellow (foredunes)	1–5 m tall; still mobile; low organic matter; harsh conditions (drought, salt spray)	Marram grass (deep roots bind sand)
3	Grey dunes	More stable; humus developing; pH falling; greater species diversity	Lichens, mosses, red fescue, sea spurge
4	Dune slack	Damp hollows between ridges; permanent water may be present	Rushes, creeping willow
5	Climax vegetation	Fully stable; rich soil; woodland develops in humid climates	Oak, ash, bramble

7. Factors Affecting Coastal Landscapes

Geology

Rock type controls how quickly coastlines erode and the shape of the coast.

- **Resistant rocks** (granite, chalk, limestone) form steep cliffs and headlands
- **Weak rocks** (clay, sands, glacial till) erode rapidly, forming bays and low cliffs

Two types of coastline structure:

- **Discordant coastline** — bands of hard and soft rock run perpendicular to the sea. Differential erosion creates alternating headlands (hard) and bays (soft). Example: Isle of Purbeck, Dorset.
- **Concordant coastline** — bands of rock run parallel to the sea. The outer band protects softer rock behind. If breached, a **cove** forms. Example: Lulworth Cove, Dorset — breach in the Purbeck limestone allowed erosion of soft Wealden Beds behind, forming the circular bay.

Swanage Bay example: Soft Wealden Beds (clays) form Swanage Bay; hard limestone/chalk at Ballard Point (east) and Durlston Head (south) resist erosion. The bay has built up a wide sandy beach. This is classic discordant coastline behaviour.

Sea Level Change

Sea level has changed significantly since the last ice age (ending ~10,000 years ago).

Eustatic Change: A global change in sea level caused by changes in the volume of water in the oceans — e.g., melting ice sheets raise sea level; glacial periods lock water as ice, lowering sea level. Currently sea level is rising due to climate change (~3.3 mm/year globally).

Isostatic Change: A local/regional change in the height of the land relative to sea level, caused by the weight of ice pressing the land down during glaciation. When the ice melts, land slowly **rebounds** (isostatic recovery). Scotland is still rising (~1 mm/year) while southern England is sinking — changing relative sea levels.

Impacts of rising sea level: increased coastal flooding, faster cliff erosion, submergence of low-lying land (e.g., Thames Estuary), saltwater intrusion into freshwater aquifers.

Ria — a drowned river valley, formed by rising sea levels after the ice age. Example: Milford Haven, Pembrokeshire. **Fjord** — a drowned glacial valley with steep sides; less common in Britain.

Human Activity

- **Hard engineering** (sea walls, groynes) interrupts natural sediment supply, causing starvation of beaches further along the coast
- **Dredging** offshore for sand and gravel reduces sediment supply to beaches
- **Urbanisation** in coastal floodplains increases flood risk
- **Climate change** (indirectly human) — rising sea levels and more intense storms

8. Coastal Management

Hard Engineering

Structure	How it works	Advantages	Disadvantages
Sea wall	Concrete/rock wall at base of cliff or top of beach reflects wave energy	Effective protection; long lifespan; allows land use behind	Very expensive (£2,000–£5,000/m); reflects wave

			energy, scours beach; ugly; requires maintenance
Revetments	Sloping wooden or rock structures absorb wave energy	Cheaper than sea walls; allows sediment through; more natural-looking	Less effective; wooden versions rot; can cause sediment starvation downdrift
Groynes	Wooden or rock structures perpendicular to shore, trap sediment moved by LSD	Build up beach width; beach protects cliff; relatively cheap (£5,000–£10,000 each)	Starve beaches downdrift of sediment; can accelerate erosion beyond groyne field
Rock armour (rip-rap)	Large boulders (granite or limestone) placed at cliff base absorb wave energy	Relatively cheap; flexible; can be natural-looking; easy to repair	Boulders often imported (not local rock); can be moved in extreme storms; not aesthetically pleasing
Offshore breakwaters	Submerged or partially submerged rock ridges break waves before they reach shore	Reduce wave energy; encourage sediment deposition in lee; less visual impact	Expensive; can be hazard to navigation; limited effectiveness in severe storms

Soft Engineering

Strategy	Description	Advantages	Disadvantages
Beach nourishment	Pumping or dumping sand/shingle onto depleted beach to widen it	Maintains beach (tourism); natural-looking; dissipates wave energy; relatively cheap short-term	Needs constant replenishment (every 5–10 years); imported sediment may differ from natural; expensive long-term
Managed retreat (coastal realignment)	Allowing the coastline to move naturally inland; breaching sea	Creates natural flood defence; encourages biodiversity; cheap;	Loss of agricultural land; compensation disputes; politically sensitive; loss of homes/businesses

	walls to allow salt marsh to develop	sustainable; supports habitats	

Exam Tip: Hard vs soft engineering often appears as a 6–8 mark evaluate question. Always link strategies to **cost, sustainability, effectiveness,** and **impact on sediment supply**. Managed retreat is increasingly favoured by the Environment Agency as a sustainable, cost-effective long-term strategy.

9. Case Study — Holderness Coast, East Yorkshire

Case Study: Holderness Coast, East Yorkshire

Context: The Holderness Coast stretches ~60 km from Flamborough Head to Spurn Head. It is the fastest-eroding coastline in Europe.

Rate of erosion: Average ~2 metres per year (up to 10 m in storms). Since Roman times, ~30 villages have been lost to the sea. The village of Skipsea has lost over 3 km of land.

Causes of rapid erosion:

- Rock type — cliffs are made of soft, easily eroded **glacial till** (boulder clay) deposited during the last ice age; no cohesion when saturated
- Long fetch — North Sea waves can have a fetch of ~500–800 km from Scandinavia, generating powerful destructive waves
- Narrow beach — beaches offer little natural protection; the beach is maintained at ~20 m width (compared to ~50 m needed for effective protection)
- Sub-aerial processes — freeze-thaw, saturation from rainfall, and slumping all weaken the cliffs before wave attack
- Groynes at Bridlington trap sediment, starving beaches to the south

Impacts:

- Houses, roads, and farmland lost — at Aldbrough, houses fall into the sea
- Gas terminal at Easington (supplies 25% of UK gas) threatened — defended with rock armour
- Caravan parks near Ulrome face loss of pitches

- Spurn Head spit threatened — storm in 2013 breached the spit, cutting off the RNLI lifeboat station

Management strategies and conflicts:

- Mappleton (1991) — rock groynes and rock armour installed at cost of ~£2 million to protect the village and the B1242 road. Successful locally but caused accelerated erosion south at Great Cowden (~2 m/year increased to ~4 m/year)
- Easington Gas Terminal — rock armour (£4.5 million). Highly justified given economic importance
- Most settlements south of Mappleton receive no protection — a politically controversial "do nothing" approach. Residents object but national policy under SMP (Shoreline Management Plan) is "no active intervention" for many sections as it is not cost-effective to defend
- Conflict: local farmers and residents vs national planners and Environment Agency — disputes over compensation

10. Case Study — Medmerry Managed Retreat, West Sussex

Case Study: Medmerry Managed Retreat, West Sussex

What happened: In 2013, the Environment Agency (EA) deliberately breached a 7 km shingle sea wall at Medmerry, near Selsey, allowing the sea to flood 183 hectares of low-lying agricultural land. This created a new, natural intertidal habitat (salt marsh, mudflats, reedbeds).

Why managed retreat?

- The old shingle bank was increasingly expensive to maintain and would have needed £30 million of hard engineering work to remain effective
- Under the EU Habitats Directive, compensatory habitat was needed to replace intertidal areas being lost to sea-level rise elsewhere
- Cost of managed retreat: ~£28 million (vs £30 million+ for hard engineering, which would have also required ongoing maintenance)

Benefits:

- 7 km of new tidal creek, salt marsh, and reedbeds created — largest intertidal habitat creation scheme in the UK

- 5,000 homes in Selsey now better protected from flooding via a new realigned earth bank further inland
- Important wildlife habitats: lapwing, ringed plover, little tern, water voles; internationally designated SSSI and SPA (Special Protection Area)
- Tourism and recreation value — new cycling/walking trails

Conflicts and limitations:

- 183 ha of agricultural land permanently lost (although farmers were compensated)
- Some local objection from landowners and farmers about land loss
- Salt marsh takes several years to fully establish

Significance: Medmerry is widely cited as a model for sustainable, cost-effective coastal management in the face of rising sea levels. It demonstrates how working with natural processes can be more effective than fighting them.

Exam Tip: For case study questions, always give specific data: rates of erosion (2 m/yr Holderness), costs (£28 million Medmerry), names of places and dates. Generic answers score poorly at GCSE. Medmerry is an excellent example for any question on managed retreat or sustainable coastal management.

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